Regression Analysis

SNU Statistics graduate entrance exam problems

53 problems arranged by topic, including 22 past exam problems (2020-10 to 2026-04 exams, restated)

Solutions are in a separate booklet.

The exam paper is given in both Korean and English; these problems follow the English version. Past exam problems are tagged [Past exam year-month #number].

Notation: unless stated otherwise, $\mathbf{y} = X\beta + \epsilon$ with X of size $n \times (p+1)$ (intercept included) and full column rank, $\epsilon \sim N_n(\mathbf{0}, \sigma^2 I)$, $\hat{\beta} = (X'X)^{-1}X'\mathbf{y}$, $H = X(X'X)^{-1}X' = (h_{ij})$, $\mathbf{e} = \mathbf{y} - X\hat{\beta}$, $SSE = \mathbf{e}'\mathbf{e}$, $MSE = SSE/(n-p-1)$, $(X'X)^{-1} = (c_{ij})$, $J = \mathbf{1}\mathbf{1}'$. In simple regression, $S_{xx} = \sum(x_i - \bar{x})^2$, $S_{xy} = \sum(x_i - \bar{x})(y_i - \bar{y})$. Upper-$\alpha$ points are $t_\alpha(\cdot)$, $F_\alpha(\cdot,\cdot)$, $\chi^2_\alpha(\cdot)$.

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A. Simple linear regression

Problem 1

Least squares in simple regression

For $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$, $i = 1, \ldots, n$:

(a) Derive $\hat{\beta}_1 = S_{xy}/S_{xx}$ and $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$, and show they minimize $\sum(y_i - \beta_0 - \beta_1 x_i)^2$.

(b) Show $\sum e_i = 0$, $\sum x_i e_i = 0$ and $\sum \hat{y}_i e_i = 0$, and deduce $SST = SSR + SSE$.

(c) Show $R^2 = r_{xy}^2$. If b is the slope of the regression of x on y, show $\hat{\beta}_1 b = r_{xy}^2$.

Problem 2

Inference in simple regression

Data: $x = (1, 2, 3, 4, 5)$, $y = (2, 4, 5, 4, 6)$.

(a) Fit the line and compute SSE, MSE and R^2.

(b) Give a 95% confidence interval for β_1 ($t_{0.025}(3) = 3.182$) and test $H_0 : \beta_1 = 0$.

(c) Give a 95% prediction interval for a new observation at $x = 6$.

(d) Show that $\mathrm{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\bar{x}\sigma^2 / S_{xx}$ and explain when the two estimates are uncorrelated.

Problem 3

Regression through the origin

Consider $y_i = \beta x_i + \epsilon_i$, $\epsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$.

(a) Find $\hat{\beta}$, its distribution, and an unbiased estimator of σ^2 (state its degrees of freedom).

(b) Show that $\sum x_i e_i = 0$ but $\sum e_i$ need not be 0.

(c) Explain why $1 - SSE/\sum(y_i - \bar{y})^2$ may be negative here, and give the appropriate decomposition of $\sum y_i^2$.

Problem 4

Gauss–Markov for the slope

In simple regression with uncorrelated errors of common variance σ^2, let $\tilde{\beta}_1 = \sum c_i y_i$ be linear and unbiased for β_1.

(a) Show that $\sum c_i = 0$ and $\sum c_i x_i = 1$.

(b) Minimize $\operatorname{Var} \tilde{\beta}_1 = \sigma^2 \sum c_i^2$ under these constraints and show the minimizer is $c_i = (x_i - \bar{x})/S_{xx}$.

(c) For a design restricted to $x_i \in [-1, 1]$ with n even, which design minimizes $\operatorname{Var} \hat{\beta}_1$?

Problem 5

Expected mean squares

In simple regression:

(a) Show $E(SSR) = \sigma^2 + \beta_1^2 S_{xx}$ and $E(SSE) = (n-2)\sigma^2$.

(b) Show that $F_0 = MSR/MSE$ equals t_0^2, where $t_0 = \hat{\beta}_1 / \sqrt{MSE/S_{xx}}$.

(c) Identify the noncentrality parameter of the distribution of SSR/σ^2.

Problem 6

Inference on correlation

(X, Y) is bivariate normal and r is the sample correlation of n pairs.

(a) Show that the t statistic for $\beta_1 = 0$ in the regression of Y on X equals $r\sqrt{n-2}/\sqrt{1-r^2}$.

(b) With $n = 28$ and $r = 0.6$, test $H_0 : \rho = 0$ at 5% ($t_{0.025}(26) = 2.056$).

(c) Using Fisher's z, give an approximate 95% confidence interval for ρ.

Problem 7

[Past exam 2025-10 #4]

Simple regression with unequal variances

$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ with independent $\epsilon_i \sim N(0, \sigma_i^2)$.

(a) If $\sigma_i^2 = \sigma^2$, find the MLEs of β_0, β_1.

(b) If $\sigma_i^2 = \sigma_1^2$ for $i \leq n_1$ and σ_2^2 for $i > n_1$, both unknown, find the MLEs of $\beta_0, \beta_1, \sigma_1^2, \sigma_2^2$.

(c) Find the LSE and the WLSE.

(d) Are the MLE and WLSE identical?

Problem 8

Lack of fit and replication

At each of m distinct x-values x_j there are n_j replicates y_{jk}, $n = \sum n_j$, and the straight-line model is fitted.

(a) Show $SSE = SS_{PE} + SS_{LOF}$, where $SS_{PE} = \sum_j \sum_k (y_{jk} - \bar{y}_j)^2$ and $SS_{LOF} = \sum_j n_j (\bar{y}_j - \hat{y}_j)^2$.

(b) Show that $F = \frac{SS_{LOF}/(m-2)}{SS_{PE}/(n-m)} \sim F(m-2, n-m)$ if the line is correct.

(c) Why can lack of fit not be tested this way without replication?

B. Least squares geometry and the hat matrix

Problem 9

The hat matrix

(a) Show that H and $I - H$ are symmetric idempotent with traces $p + 1$ and $n - p - 1$.

(b) Show $\mathbf{e} = (I - H)\epsilon$, $\operatorname{Var} \mathbf{e} = \sigma^2(I - H)$ and $\operatorname{Cov}(\hat{\beta}, \mathbf{e}) = 0$.

(c) Show $\frac{1}{n} \leq h_{ii} \leq 1$ (with an intercept) and that $h_{ii} = 1$ implies $e_i = 0$.

(d) In simple regression show $h_{ii} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}}$.

Problem 10

Quadratic forms

Let $\mathbf{y} \sim N(\mu, \Sigma)$ and A symmetric.

(a) Show $E(\mathbf{y}'A\mathbf{y}) = \text{tr}(A\Sigma) + \mu' A\mu$.

(b) Use (a) to compute $E(SST)$, $E(SSR)$ and $E(SSE)$ in the multiple regression model.

(c) Show that $\mathbf{y}'(H - J/n)\mathbf{y}$ and $\mathbf{y}'(I - H)\mathbf{y}$ are independent and give their distributions.

Problem 11

[Past exam 2021-04 #1]

Centred regression and SSR

Centred regression $y_i = \sum_j \beta_j x_{ij} + \epsilon_i$ with $\sum_i y_i = \sum_i x_{ij} = 0$, $n > p$, X of full rank.

(a) Write $SSR = \sum \hat{y}_i^2$ and $SST = \sum y_i^2$ in terms of X and $\mathbf{y}$.

(b) Prove $\max_{\beta \neq \mathbf{0}} \dfrac{\langle X\beta, \mathbf{y} \rangle^2}{\|X\beta\|^2 \|\mathbf{y}\|^2} = \dfrac{SSR}{SST}$.

(c) Find the maximizer.

Problem 12

[Past exam 2022-04 #4]

SVD and eigendecomposition

Define the singular value decomposition of an $n \times p$ matrix X $(n > p)$ and the eigendecomposition of $X'X$, and relate them.

Problem 13

Partial regression

Let $X = (X_1, X_2)$ and $H_1 = X_1(X_1'X_1)^{-1}X_1'$.

(a) Show $\hat{\beta}_2 = \{X_2'(I - H_1)X_2\}^{-1}X_2'(I - H_1)\mathbf{y}$.

(b) Show that the residuals from regressing $(I - H_1)\mathbf{y}$ on $(I - H_1)X_2$ equal the full-model residuals.

(c) If $X_1'X_2 = 0$, show that $\hat{\beta}_1$ is the same in the full model and in the model with X_1 alone, and that SSR splits into two parts.

Problem 14 **[Past exam 2022-04 #6]**

Frisch–Waugh–Lovell

Model $y_j = \mathbf{x}_{1j}'\beta_1 + \mathbf{x}_{2j}'\beta_2 + \epsilon_j$ ($\mathbf{x}_1$ includes the intercept). Regress y on $\mathbf{x}_1$ to get residuals e_j; regress each component of $\mathbf{x}_2$ on $\mathbf{x}_1$ to get residuals $\mathbf{x}_{2\cdot 1,j}$; regress e_j on $\mathbf{x}_{2\cdot 1,j}$ to get $\tilde{\beta}_2$.

(i) Prove that $\tilde{\beta}_2 = \hat{\beta}_2$ of the full model.

(ii) Interpret.

Problem 15

[Past exam 2020-10 #5]

Projections and partial regression

$X = (X_1, X_2)$ of full rank ($n \times q$ and $n \times (p-q)$), H, H_1, H_2 the projections onto the column spaces of X, X_1, X_2, and $E(\mathbf{y} \mid X) = X_1\beta_1 + X_2\beta_2$.

(a) Express $\hat{\beta}_2$ using H_1, X_2, $\mathbf{y}$.

(b) Find $H\mathbf{y} - H_1\mathbf{y}$ when $X_1'X_2 = 0$.

Problem 16

[Past exam 2023-10 #4]

Full versus reduced model estimators

With $X = (X_k, X_r)$, let $\hat{\beta}_k$ be the part of the full least squares estimator and $\tilde{\beta}_k = (X_k'X_k)^{-1}X_k'\mathbf{y}$ the reduced-model estimator. Prove that $\mathrm{Var}(\hat{\beta}_k) - \mathrm{Var}(\tilde{\beta}_k)$ is nonnegative definite.

Problem 17

[Past exam 2024-10 #4]

Properties of least squares fits

$(\mathbf{x}_i, y_i) \in \mathbb{R}^p \times \mathbb{R}$, $\sum \mathbf{x}_i \mathbf{x}_i'$ positive definite, $\hat{\beta}$ the least squares estimate.

(a) If $y_i = x_{ij}$ for all i and a fixed j, find $\hat{\beta}$.

(b) Assuming $\sum y_i^2 > 0$, prove $\max_\beta \dfrac{(\sum y_i \mathbf{x}_i'\beta)^2}{\sum y_i^2 \sum (\mathbf{x}_i'\beta)^2} = \dfrac{\sum \hat{y}_i^2}{\sum y_i^2}$, and evaluate it in case (a).

Problem 18

[Past exam 2025-04 #4]

Residuals and fitted values

For $\mathbf{y} = X\beta + \epsilon$ with k regressors and intercept:

(1) find $E(\mathbf{e}'\mathbf{e})$ and $\mathrm{Cov}(\hat{\beta}, \mathbf{e})$;

(2) show $\sum_i \mathrm{Var}(\hat{y}_i) = c\sigma^2$ and find c;

(3) show $H - \mathbf{1}(\mathbf{1}'\mathbf{1})^{-1}\mathbf{1}'$ is idempotent.

C. Inference in the linear model

Problem 19

Overall F test

(a) Show that $F_0 = MSR/MSE$ has the noncentral distribution $F(p, n-p-1, \lambda)$ with $\lambda = \beta' X'(I - J/n)X\beta/(2\sigma^2)$.

(b) Show that $\lambda = 0$ iff $\beta_1 = \cdots = \beta_p = 0$.

(c) Express F_0 in terms of R^2.

Problem 20

[Past exam 2025-04 #5]

R^2 of nested models

$\mathbf{y} = X_1\beta_1 + X_2\beta_2 + \epsilon$. Show that $R^2(F) \geq R^2(R)$, where R is the reduced model under $H_0 : \beta_2 = \mathbf{0}$.

Problem 21

Partial and sequential F

(a) Show $SS(\hat{\beta}_j \mid \text{all others}) = \hat{\beta}_j^2 / c_{jj}$, so the partial F test equals the squared t test.

(b) Show that the sequential sums of squares $SS(\hat{\beta}_1 \mid \hat{\beta}_0)$, $SS(\hat{\beta}_2 \mid \hat{\beta}_0, \hat{\beta}_1)$, $\ldots$ add up to SSR.

(c) Show that, if the centred regressors are mutually orthogonal, the sequential sums of squares do not depend on the order.

Problem 22

Restricted least squares

Data with X having rows $(1, x_{i1}, x_{i2})$:

x_1	1	2	3	4	5	6
x_2	0	1	1	3	2	4
y	3	6	6	10	9	13

Here $X'X = \begin{pmatrix} 6 & 21 & 11 \\ 21 & 91 & 51 \\ 11 & 51 & 31 \end{pmatrix}$, $(X'X)^{-1} = \frac{1}{200}\begin{pmatrix} 220 & -90 & 70 \\ -90 & 65 & -75 \\ 70 & -75 & 105 \end{pmatrix}$, $X'\mathbf{y} = (47, 196, 112)'$.

(a) Compute $\hat{\beta}$ and SSE.

(b) Test $H_0 : \beta_1 = \beta_2$ at 5% by the formula $Q = (C\hat{\beta})'\{C(X'X)^{-1}C'\}^{-1}C\hat{\beta}$ ($F_{0.05}(1,3) = 10.13$).

(c) Verify the result by fitting the reduced model.

Problem 23

[Past exam 2026-04 #4]

Constrained least squares

For $\mathbf{y} = X\beta + \epsilon$ and full-rank C $((p+1) \times q)$:

(a) sketch why the minimizer of $\|\mathbf{y} - X\beta\|^2$ subject to $C'\beta = \mathbf{0}$ is $\hat{\beta}_r = \hat{\beta} - (X'X)^{-1}C\{C'(X'X)^{-1}C\}^{-1}C'\hat{\beta}$;

(b) describe the partial F test of $C'\beta = \mathbf{0}$ and its null distribution;

(c) derive the distribution of $C'\hat{\beta}$ and the corresponding F test;

(d) are the tests in (b) and (c) equivalent?

Problem 24

[Past exam 2023-10 #5]

Least squares under a linear constraint

$y_i = \theta_i + \epsilon_i$, $i = 1, \ldots, 4$, $\epsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$, with the constraint $\theta_1 + \theta_2 + \theta_3 + \theta_4 = 0$. Show that the F statistic for $H_0 : \theta_1 = \theta_3$ is $F_0 = \dfrac{2(y_1 - y_3)^2}{(y_1 + y_2 + y_3 + y_4)^2}$.

Problem 25

Simultaneous inference

(a) Show that $(\hat{\beta}-\beta)'X'X(\hat{\beta}-\beta)/\{(p+1)MSE\} \sim F(p+1, n-p-1)$.

(b) Derive Scheffé's simultaneous intervals for all $\mathbf{a}'\beta$ from (a).

(c) For $p+1=3$ and $n-p-1=20$, compare the multipliers of single, Bonferroni and Scheffé intervals for the three coefficients ($t_{0.025}(20) = 2.086$, $t_{0.05/6}(20) = 2.613$, $F_{0.05}(3, 20) = 3.098$).

Problem 26

Comparing two regressions

Two groups follow $y_{ij} = \beta_{0i} + \beta_{1i} x_{ij} + \epsilon_{ij}$, $i = 1, 2$, $j = 1, \ldots, n_i$, with a common σ^2.

(a) Write a single regression with an indicator d (for group 2) and its interaction with x that reproduces the two separate lines.

(b) Show that its SSE equals $SSE_1 + SSE_2$ of the separate fits.

(c) Give the F tests for identical lines and for parallel lines, with degrees of freedom.

Problem 27

Indicator variables

(a) A factor has three levels. Explain why using three indicators plus an intercept makes $X'X$ singular.

(b) With baseline coding, interpret the coefficients of $E(y) = \beta_0 + \beta_1 x + \beta_2 d_1 + \beta_3 d_2$.

(c) How would you test that the three levels have the same intercept? Give the degrees of freedom.

Problem 28

Piecewise linear regression

Consider $E(y) = \beta_0 + \beta_1 x + \beta_2 (x - c)_+$ with known c.

(a) Show that the mean function is continuous at c and find the slopes on each side.

(b) Modify the model to allow a jump at c.

(c) How would you test whether the slope changes at c?

Problem 29

[Past exam 2024-05 #5]

Two-way ANOVA without replication

Two-way ANOVA without replication, $y_{ij} = \mu + \alpha_i + \beta_j + \epsilon_{ij}$, $i \le a$, $j \le b$, $\sum \alpha_i = \sum \beta_j = 0$, $\epsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2)$.

(1) Prove that $Q_1 = \sum_{i,j} (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2$ and $Q_2 = \sum_{i,j} (y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot})^2$ are independent.

(2) Find an unbiased estimator of σ^2 and a $100(1-\alpha)\%$ confidence interval for $\mu + \alpha_1 + \beta_1$.

D. Diagnostics and model selection

Problem 30

Deletion formulas

Let $\hat{\beta}_{(i)}$ and $s^2_{(i)}$ be computed without observation i.

(a) Using $(A-\mathbf{x}\mathbf{x}')^{-1} = A^{-1}+A^{-1}\mathbf{x}\mathbf{x}'A^{-1}/(1-\mathbf{x}'A^{-1}\mathbf{x})$, prove $\hat{\beta}_{(i)} = \hat{\beta}-(X'X)^{-1}\mathbf{x}_i e_i/(1-h_{ii})$.

(b) Prove $(n-p-2)s^2_{(i)} = (n-p-1)s^2 - e_i^2/(1-h_{ii})$.

(c) Show Cook's distance satisfies $D_i = \frac{r_i^2}{p+1}\cdot\frac{h_{ii}}{1-h_{ii}}$, with $r_i = e_i/(s\sqrt{1-h_{ii}})$.

Problem 31

Studentized residuals and PRESS

(a) Show that $r_i^* = e_i/(s_{(i)}\sqrt{1-h_{ii}}) \sim t(n-p-2)$.

(b) Show $r_i^* = r_i\sqrt{(n-p-2)/(n-p-1-r_i^2)}$, and deduce that $r_i^2 \leq n-p-1$.

(c) Show that the leave-one-out cross-validation error is PRESS $= \sum_i \{e_i/(1-h_{ii})\}^2$.

Problem 32

[Past exam 2026-04 #5]

Studentized residuals

(a) Show $e_i/(\sigma\sqrt{1-h_{ii}}) \sim N(0,1)$.

(b) Why is $r_i = e_i/(s\sqrt{1-h_{ii}})$ not t-distributed? (c) With $s_{(i)}$ computed without observation i and $r_i^* = e_i/(s_{(i)}\sqrt{1-h_{ii}})$, show (i) $s_{(i)}^2 = \{(n-p-1)s^2 - e_i^2/(1-h_{ii})\}/(n-p-2)$ and (ii) $\hat{\beta}_{(i)} = \hat{\beta} - (X'X)^{-1}\mathbf{x}_i e_i/(1-h_{ii})$.

Problem 33

[Past exam 2022-04 #5]

Leverage, outliers and influence

Describe how to diagnose high-leverage points, outliers and influential points.

Problem 34

Selection criteria

(a) Show that R^2 cannot decrease when a regressor is added.

(b) Show that R_a^2 increases when a regressor is added iff its partial F statistic exceeds 1.

(c) Show that $E(C_k) \approx k + 1$ when the k-variable model is correct, and that $C_p = p + 1$ for the full model.

Problem 35

[Past exam 2025-04 #6]

Mallows' C_p

For a submodel with p parameters, $C_p = SSE(p)/\hat{\sigma}^2 - n + 2p$, $SSE(p) = \mathbf{y}'(I - H_p)\mathbf{y}$. Show $E(C_p) \approx p$ if $\mu = E\mathbf{y} = X_p\beta_p$.

Problem 36

[Past exam 2021-10 #1]

Out-of-sample prediction error

$y_i = \mathbf{x}_i'\beta + e_i$, $i = 1, \ldots, n$, independent, $E(e_i \mid \mathbf{x}_i) = 0$, $\mathrm{Var}(e_i \mid \mathbf{x}_i) = \sigma^2$, and $\hat{\beta}$ is fitted on the first n_1 observations only.

(1) Express $E\frac{1}{n-n_1}\sum_{i>n_1}(y_i - \mathbf{x}_i'\hat{\beta})^2$ in terms of σ^2 and the $\mathbf{x}_i$.

(2) Is the training error $\frac{1}{n_1}\sum_{i\le n_1}(y_i - \mathbf{x}_i'\hat{\beta})^2$ an unbiased estimator of it?

E. Departures from assumptions and remedies

Problem 37

Transformations

(a) Show that $(y^\lambda - 1)/\lambda \to \log y$ as $\lambda \to 0$.

(b) Suppose $\mathrm{Var}(y) = g(\mu)$ with $\mu = Ey$. Using a first-order Taylor expansion, find h such that $\mathrm{Var}\, h(y)$ is approximately constant when $g(\mu) = \mu$ and when $g(\mu) = \mu^2$.

(c) Explain why, in the Box–Cox procedure, the residual sums of squares are computed for $z^{(\lambda)} = (y^\lambda - 1)/(\lambda \dot{y}^{\lambda-1})$ ($\dot{y}$ the geometric mean) rather than for $y^{(\lambda)}$.

Problem 38

Errors in variables

The true model is $y = \beta_0 + \beta_1 x^* + \epsilon$, but $x = x^* + \delta$ is observed, where x^*, δ, ϵ are independent with $\operatorname{Var} x^* = \sigma_*^2$, $\operatorname{Var} \delta = \sigma_\delta^2$.

(a) Find the probability limit of the least squares slope of y on x.

(b) If σ_δ^2 is known (from replicate measurements), propose a consistent estimator of β_1.

(c) In Berkson's model, $x^* = x + \delta$ with x set by the experimenter and δ independent of x. Is least squares unbiased?

Problem 39

Omitted variables

The true model is $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon$, but y is regressed on x_1 alone.

(a) Show $E\tilde{\beta}_1 = \beta_1 + \beta_2 S_{12}/S_{11}$, with $S_{12} = \sum(x_{i1} - \bar{x}_1)(x_{i2} - \bar{x}_2)$.

(b) Compare $\operatorname{Var}\tilde{\beta}_1$ with $\operatorname{Var}\hat{\beta}_1$ in the full model, in terms of the sample correlation r_{12}.

(c) Show that $\operatorname{MSE}(\tilde{\beta}_1) \leq \operatorname{Var}(\hat{\beta}_1)$ iff $\beta_2^2 \leq \operatorname{Var}(\hat{\beta}_2)$.

Problem 40

Weighted least squares

Consider $y_i = \beta x_i + \epsilon_i$ with $x_i > 0$ and independent $\epsilon_i \sim N(0, \sigma^2 x_i)$.

(a) Find the GLS estimator of β and its variance.

(b) Find the variance of the OLS estimator $\sum x_i y_i / \sum x_i^2$ and verify that it is not smaller.

(c) Give an unbiased estimator of σ^2.

Problem 41

When OLS is GLS

Let $\operatorname{Var}\epsilon = \sigma^2 V$ with $V = (1-\rho)I + \rho J$, $-\frac{1}{n-1} < \rho < 1$, and let X contain the intercept column.

(a) Show that $V^{-1} = \frac{1}{1-\rho}\big(I - \frac{\rho}{1-\rho+n\rho}J\big)$.

(b) Show that the GLS and OLS estimators of β coincide.

(c) Is the usual MSE unbiased for σ^2 in this model?

Problem 42

[Past exam 2020-10 #4]

GLS is BLUE

$\mathbf{y} \sim N(X\beta, V)$ with known nonsingular V. Let $\hat{\beta} = (X'V^{-1}X)^{-1}X'V^{-1}\mathbf{y}$, and let $\tilde{\beta} = A\mathbf{y}$ be any linear unbiased estimator. Show $\mathbf{a}'\{\mathrm{Var}\,\tilde{\beta} - \mathrm{Var}\,\hat{\beta}\}\mathbf{a} \geq 0$ for all $\mathbf{a}$.

Problem 43

[Past exam 2023-04 #2]

Generalized least squares

$\mathbf{y} = X\beta + \epsilon$, $E\epsilon = \mathbf{0}$, $\operatorname{Var}\epsilon = \sigma^2 V$ with V known positive definite, X of full rank $k+1$.

(1) Find the mean and covariance of $\tilde{\beta} = (X'X)^{-1}X'\mathbf{y}$.

(2) Show that the BLUE is $\hat{\beta} = (X'V^{-1}X)^{-1}X'V^{-1}\mathbf{y}$.

(3) Find $\operatorname{Var}\hat{\beta}$.

(4) Give an unbiased estimator of σ^2.

(5) How can V be estimated when unknown?

Problem 44

Variance inflation

(a) Show that $c_{jj} = (1 - R_j^2)^{-1}$ for standardized regressors, where R_j^2 is from regressing x_j on the others.

(b) Show $\sum_j \mathrm{VIF}_j = \sum_j 1/\lambda_j$, with λ_j the eigenvalues of $X'X$.

(c) For two regressors with correlation 0.95, find the VIFs and the factor by which confidence intervals widen.

Problem 45

Ridge regression

Let $\hat{\beta}(k) = (X'X + kI)^{-1}X'\mathbf{y}$, with X standardized and $X'X = P\Lambda P'$.

(a) Show that $\hat{\beta}(k)$ minimizes $\|\mathbf{y} - X\beta\|^2 + k\|\beta\|^2$ and is the OLS estimator for the augmented data $\binom{\mathbf{y}}{\mathbf{0}}$, $\binom{X}{\sqrt{k}I}$.

(b) Find $E\hat{\beta}(k)$ and $\operatorname{Var}\hat{\beta}(k)$.

(c) Show that $E\|\hat{\beta}(k) - \beta\|^2 < E\|\hat{\beta} - \beta\|^2$ for some $k > 0$.

(d) If $X'X = I$, find the k minimizing $E\|\hat{\beta}(k) - \beta\|^2$.

Problem 46

[Past exam 2021-04 #2]

Ridge regression with rank deficiency

In the same centred model with uncorrelated errors of variance σ^2 but X possibly rank deficient, let $\hat{\beta}_{\text{ridge}}$ minimize $\|\mathbf{y} - X\beta\|^2 + k\|\beta\|^2$, $k > 0$.

(a) Find it.

(b) With λ_i the eigenvalues of $X'X$, prove $\sum_j \operatorname{Var} \hat{\beta}_{j,\text{ridge}} = \sigma^2 \sum_i \lambda_i/(\lambda_i + k)^2$.

Problem 47

[Past exam 2021-10 #2]

Ridge regression via augmented data

$E(\mathbf{y} \mid X) = X\beta$, $\mathrm{Var}(\mathbf{y} \mid X) = \sigma^2 I$, $X = UDV'$ (thin SVD). With $X^* = \binom{X}{\sqrt{\lambda} I_p}$ and $\mathbf{y}^* = \binom{\mathbf{y}}{\mathbf{0}_p}$, let $\hat{\beta} = (X^{*\prime} X^*)^{-1} X^{*\prime} \mathbf{y}^*$. Express $E(\hat{\beta} \mid X)$ and $\mathrm{Var}(\hat{\beta} \mid X)$ in terms of U, D, V, β, σ.

Problem 48

[Past exam 2025-10 #5]

Ridge versus least squares

Ridge $\hat{\beta}_R = (X'X + \lambda I)^{-1} X'\mathbf{y}$.

(a) Find its mean and covariance.

(b) Show that for some λ its variance is smaller than that of the LSE.

(c) Find and interpret the limits as $\lambda \to 0$ and $\lambda \to \infty$.

Problem 49

Principal component regression

With $X'X = \sum_k l_k \mathbf{e}_k \mathbf{e}_k'$ ($l_1 \geq \cdots \geq l_p$), the PCR estimator with r components is

$$\tilde{\beta}_r = \sum_{k \leq r} l_k^{-1} \mathbf{e}_k \mathbf{e}_k' X' \mathbf{y}.$$

(a) Find the bias and covariance of $\tilde{\beta}_r$.

(b) Show $E\|\tilde{\beta}_r - \beta\|^2 < E\|\hat{\beta} - \beta\|^2$ iff $\sum_{k>r} (\mathbf{e}_k'\beta)^2 < \sigma^2 \sum_{k>r} 1/l_k$.

(c) Contrast PCR with ridge in terms of the shrinkage factors applied to the canonical coefficients.

Problem 50

[Past exam 2024-10 #5]

Subgradients and penalized estimation

For convex $L : \mathbb{R} \to \mathbb{R}$, $\partial L(\theta) = \{z : L(\phi) \geq L(\theta) + z(\phi - \theta)\ \forall \phi\}$.

(a) Prove θ minimizes L iff $0 \in \partial L(\theta)$.

(b) Show $\partial|\theta| = \{\operatorname{sgn}\theta\}$ for $\theta \neq 0$ and $[-1, 1]$ at 0. (c), (d) For centred data with $\sum x_i^2 > 0$, let $L_\lambda(\theta) = \frac{1}{2}\sum(y_i - x_i\theta)^2 + \lambda|\theta|$ and θ_0 the least squares slope. Prove the minimizer is $\hat{\theta}_\lambda = \operatorname{sgn}(\theta_0)\big(|\theta_0| - \lambda/\sum x_i^2\big)_+$. Is it unique?

F. Logistic regression

Problem 51

Logistic regression: estimation

For $P(y_i = 1) = \pi_i = e^{\mathbf{x}_i'\beta}/(1 + e^{\mathbf{x}_i'\beta})$:

(a) Derive the score and Hessian, and show that the log-likelihood is concave.

(b) Show that a Newton–Raphson step is a weighted least squares regression of a working response.

(c) Interpret e^{β_j}.

Problem 52

Logistic regression: 2×2 *tables*

(a) For a binary x, show that the MLEs of the logistic model give $e^{\hat{\beta}_1} = \frac{n_{11}n_{00}}{n_{10}n_{01}}$.

(b) Show that the odds ratio is the same whether computed from $P(y \mid x)$ or $P(x \mid y)$, and explain why this matters for case–control studies.

(c) With $n_{11} = 20, n_{10} = 80, n_{01} = 10, n_{00} = 90$, compute $\widehat{OR}$ and an approximate 95% interval, using $\mathrm{se}(\log \widehat{OR}) = \sqrt{\sum 1/n_{xy}}$.

Problem 53

Separation

Data: $x = (1, 2, 3, 4)$, $y = (0, 0, 1, 1)$, with model $\text{logit}\, P(y = 1) = \beta_0 + \beta_1 x$.

(a) Show that the log-likelihood along $\beta = t(-2.5, 1)$ increases to 0 as $t \to \infty$.

(b) Conclude that the MLE does not exist.

(c) Can this happen in ordinary least squares?